AI-Assisted Proposal Triage Process Flow (MATACS-TH-2026-02)

End-to-End Architecture: Dynamic Admin Config → PDF Ingestion → Schema-Bound AI Extraction → Deterministic Rule Gates → Scoring & Tie-Breakers → Human Audit

- Admin / Reviewer Input
- AI / LLM Step (Extraction & 1-Line Reason Only)
- Deterministic Code / Rules (Gates, Math & Sorting)
- Flagged Queue (Human Exception Review)
- Ineligible (Mandatory Gate Rejection)
- Human-in-the-Loop Checkpoint & Audit Log

STAGE 0 & 1 — CONFIGURATION & DOCUMENT INGESTION

PLAIN CODE + HUMAN SETUP

ADMIN SCREEN (HUMAN)

Dynamic Category Config (v1..vN)

- Category: Commercial Solar (150 kW)
- Hard Caps: ≤ $260k CAD | ≤ 16 Weeks
- Mandatory: Ontario Licence + WSIB/$5M
- Weights: Exp 30%, Price 25%, Time 15%, Warr 15%, Team 15%

REVIEWER ACTION

Batch PDF Upload (Proposals A–E)

- Selects active Category version
- Uploads vendor PDF files
- Computes SHA-256 hash per file for caching

DETERMINISTIC CODE

Stage 1: PDF Parser & Sanitizer

- Extracts text + page/line coordinate map
- Detects invisible/0pt font or macro exploits
- Masks vendor names (optional anti-bias mode)

▼ Passes + Admin Criteria JSON Schema ▼

STAGE 2 — SCHEMA-CONSTRAINED SEMANTIC EXTRACTION

AI / LLM LAYER (TEMPERATURE = 0)

AI / LLM EXTRACTION ENGINE

Structured Fact & Citation Extractor

- Maps unstructured prose into strict JSON schema
- Extracts raw values + verbatim `source_quote` & page
- Outputs `null` or `"unclear"` if missing (never guesses)
- Scans for embedded grading commands (`injection_detected`)
- **Does NOT score, rank, or perform math**

DETERMINISTIC CODE

Anti-Hallucination Quote Verifier

- Validates JSON schema types & required keys
- Checks verbatim `source_quote` exists in PDF text
- Caches verified extraction by (`file_hash`, `criteria_v`)

▼ Verified Structured JSON Enters Sequential Rule Gates ▼

STAGE 3 — SEQUENTIAL PASS/FAIL & INTEGRITY RULE GATES

100% DETERMINISTIC CODE

GATE 0: SECURITY & INTEGRITY

Prompt Injection & Anomaly Gate

- Checks `injection_detected == false`
- Checks timeframe clarity (e.g., 2021–2026 vs "since 2020")
- **Catches Proposal C** (Appendix C Page 9 directive)

GATE 1: MANDATORY CRITERIA

Pass/Fail Dealbreaker Gate

- `licence_held_by_bidder === true` (ECRA/ESA #)
- `wsib_confirmed === true` & `insurance ≥ $5M`
- **Rejects Proposal B** (Unconfirmed subcontractor)

GATE 2: PROJECT CONSTRAINTS

Budget Ceiling & Timeline Gate

- Checks `price_cad ≤ $260,000`
- Checks `timeline_weeks != null` & `≤ 16 wks`
- **Catches Proposal D** (Schedule "TBD after site assessment")

OUTPUT STATUS: FLAGGED

Hold for Human Triage (Proposals C & D)

- `rank: null, score: null`
- **Prop C:** Prompt injection on p.9 + "since 2020" ambiguity
- **Prop D:** Missing completion timeline (cannot verify ≤16w)

OUTPUT STATUS: INELIGIBLE

Disqualified (Proposal B)

- `rank: null, score: null`
- **Prop B:** Fails mandatory Ontario electrical licence rule (subcontracted post-award)

▼ Eligible & Complete Proposals (Proposals A & E) Proceed to Scoring ▼

STAGE 4 — WEIGHTED SCORING, TIE-BREAKING & SUMMARY REASON

DETERMINISTIC MATH + RESTRICTED AI SUMMARY

STAGE 4A: DETERMINISTIC MATH

Weighted Rubric Calculator (0–100)

- **Experience (30%):** Prop A (14) > Prop E (8)
- **Price (25%):** Prop A ($238k) > Prop E ($242.5k)
- **Timeline (15%):** Prop A (12w) > Prop E (13w)
- **Warranty (15%):** Prop E (25y/12y) > Prop A (25y/10y)
- **Team/Approach (15%):** Both 4/4 items present

STAGE 4B: DETERMINISTIC SORT

4-Tier Tie-Breaker Hierarchy

- Triggered if `round(Score_X, 2) == round(Score_Y, 2)`
- 1. Higher Relevant Experience install count (30% wt)
- 2. Lower Total Price CAD (25% wt)
- 3. Longer Workmanship Warranty years
- 4. Co-rank + flag for human tie arbitration

STAGE 4C: AI / TEMPLATE FORMATTER

1-Line Explainable Reason Generator

- Converts verified scorecard facts into concise 1-line `reason`
- Cannot alter numeric scores, statuses, or ranks
- Produces final `ranking.json` payload

▼ Outputs ranking.json + Full Citation Breakdown to Reviewer UI ▼

STAGE 5 — HUMAN-IN-THE-LOOP REVIEW, OVERRIDE & IMMUTABLE AUDIT LOG

HUMAN DECISION MAKER

HUMAN CHECKPOINT

Reviewer Triage Dashboard & Overrides

- **Ranked:** #1 Prop A (Score: 93), #2 Prop E (Score: 82)
- **Flagged:** Inspect Prop C (security alert) & Prop D (request schedule)
- **Override:** Any manual rank/status edit requires written justification

COMPLIANCE & VERSIONING

Immutable Audit Log & Final Award

- Logs `file_hash`, `criteria_v1`, `model_id`, quotes, scores
- If Admin edits criteria to `v2`, marks `v1` scores as *Stale*
- Human clicks **Confirm Final Award Decision**